

Cardiomyopathy Dilated 2H (CMD2H) — Comprehensive Disease Report

Disease: Cardiomyopathy, Dilated, 2H (CMD2H) **MONDO ID:** MONDO:0859358 **OMIM (phenotype):** 620203 **Causal gene:** GET3 / ASNA1 (TRC40), 19p13.13 **Category:** Mendelian (autosomal recessive)

Summary

Cardiomyopathy Dilated 2H (CMD2H; MONDO:0859358, OMIM 620203) is an ultra-rare, autosomal recessive, infantile-onset, rapidly progressive dilated cardiomyopathy caused by biallelic loss-of-function variants in GET3/ASNA1 (TRC40; NCBI Gene 439; 19p13.13). GET3/ASNA1 encodes the cytosolic ATPase that recognizes the transmembrane domain of tail-anchored (TA) membrane proteins and delivers them to the endoplasmic reticulum (ER) for post-translational insertion via the GET/TRC pathway. Reduced GET3 function impairs TA-protein insertion and cardiomyocyte proteostasis, producing ventricular dilation, heart failure, and death in early infancy.

A critical disambiguation underpins this report. The disease name "Cardiomyopathy Dilated 2H" and MONDO:0859358 map to OMIM 620203 and are caused by **GET3/ASNA1**, not by GATAD1. GATAD1 causes a *different* recessive dilated cardiomyopathy, **CMD2B** (OMIM 614672). Early iterations of this investigation initially followed the GATAD1 literature; identifier cross-referencing through Monarch/OLS4 established that MONDO:0859358 → OMIM:620203 → GET3/ASNA1, and the report was corrected accordingly. GATAD1/CMD2B findings are retained below only as an explicitly labeled contrast, to prevent future confusion, and should **not** be attributed to CMD2H.

The molecular and clinical picture of CMD2H is supported by a coherent evidence chain: the index human family (compound heterozygous ASNA1 variants p.Gln305 and p.Val163Ala in two siblings with fatal infantile DCM; Verhagen et al. 2019), an *in vitro* TA-protein insertion assay demonstrating reduced function of the missense allele, zebrafish rescue experiments confirming both variants are deleterious, and cardiomyocyte-specific Asna1*-knockout mice (Feng et al.

2025) that recapitulate ventricular dilation, myocardial thinning, and disrupted TA-protein proteostasis. There is currently no disease-specific or targeted therapy; management follows standard guideline-directed heart-failure care, with transplantation as the definitive option.

Section 1 — Disease Information

Overview. CMD2H is a Mendelian, autosomal recessive form of dilated cardiomyopathy (DCM) — a cardiac muscle disorder defined by left-ventricular (or biventricular) dilation and systolic dysfunction in the absence of abnormal loading conditions or coronary disease sufficient to explain it. CMD2H is distinguished from common idiopathic DCM by its monogenic recessive etiology (biallelic GET3/ASNA1 variants), its very early (infantile) onset, and its rapid, frequently fatal course.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0859358 ("cardiomyopathy, dilated, 2H")
OMIM (phenotype)	620203
OMIM (gene, GET3/ASNA1)	601913
MedGen	C5774296
UMLS	C5774296
GARD	0026714
NCBI Gene	439 (GET3; aliases ASNA1, TRC40, ASNA-I, ARSA1, CMD2H)
Ensembl (gene)	ENSG00000198356
HGNC symbol	GET3 (previously ASNA1)

Cross-references were verified via Monarch and OLS4. Note that "CMD2H" is itself listed among the official gene aliases for GET3 in NCBI Gene.

Synonyms / alternative names. Dilated cardiomyopathy 2H; DCM 2H; CMD2H; (gene-based) ASNA1-related / GET3-related dilated cardiomyopathy; ASNA1-related infantile dilated cardiomyopathy.

Information source. Disease-level, aggregated Mendelian resources (OMIM, MONDO, MedGen) combined with a small number of primary case reports and functional studies. The clinical description derives from a single index family (two siblings) plus supporting model-organism and in vitro data — it is *not* derived from large EHR cohorts, reflecting the ultra-rare nature of the disorder.

Section 2 — Etiology

Primary cause: genetic. CMD2H is caused by **biallelic (compound heterozygous or homozygous) loss-of-function variants in GET3/ASNA1**. It is a monogenic, autosomal recessive disorder with no established environmental or infectious primary cause.

"We identified compound heterozygous variants in the highly conserved ASNA1 gene (arsA arsenite transporter, ATP-binding, homolog), which encodes an ATPase required for post-translational membrane insertion of tail-anchored proteins." — Verhagen et al. 2019, [PMID: 31461301](#)

"Biallelic variants in ASNA1 cause severe pediatric cardiomyopathy and early death." — Verhagen et al. 2019, [PMID: 31461301](#)

Genetic risk factors. The causal risk factor is inheritance of two damaging GET3/ASNA1 alleles. Consistent with a recessive mechanism, gnomAD shows GET3/ASNA1 is constrained but tolerant of *heterozygous* loss of function (pLI = 0.05; LoF observed/expected = 15/32.4; oe_lof = 0.46, 90% CI 0.31–0.71; lof_z = 2.59; mis_z = 2.51). This constraint profile — tolerant of one null allele but depleted overall — is exactly what is expected for a gene in which heterozygous carriers are healthy and disease requires biallelic hits.

Environmental risk / protective factors / gene–environment interactions. No environmental risk factors, protective factors, or gene–environment interactions specific to CMD2H have been reported. Given the severe, near-fully-penetrant recessive genotype, environmental modifiers are unlikely to be dominant contributors, though data are too sparse to exclude modifiers of severity. (For DCM broadly, "second-hit" models — inflammation, clonal hematopoiesis, environmental triggers layered on genetic susceptibility — are increasingly recognized [PMID: 41898170], but these apply to common/idiopathic DCM, not specifically CMD2H.)

Section 3 — Phenotypes

The CMD2H phenotype, as documented in the index sibling pair, is a **severe, rapidly progressive, infantile-onset dilated cardiomyopathy**:

Phenotype	Type	Onset	Severity / course	HPO suggestion
Dilated cardiomyopathy	Clinical sign (imaging)	Infantile / early	Severe, rapidly progressive	HP:0001644
Ventricular systolic dysfunction	Clinical sign	Infantile	Severe	HP:0001635 (congestive heart failure)
Ventricular dilation	Structural	Infantile	Severe	HP:0001711 (abnormal LV morphology)
Heart failure	Clinical syndrome	Infantile	Severe, fatal	HP:0001635
Death in infancy	Outcome	Early infancy	Fatal	HP:0001522

Phenotype characteristics. Age of onset: neonatal/infantile. Severity: severe. Progression: rapidly progressive. Frequency among affected individuals: in the reported family, both biallelic siblings were affected and died in early infancy (high penetrance for the severe biallelic genotype), though the total number of reported patients is very small ($n \approx 2$ index cases), limiting frequency estimates.

Quality-of-life impact. Not formally measured with instruments (EQ-5D, SF-36, PROMIS) given infantile lethality. The practical impact is maximal — the disease is life-limiting in infancy in reported cases.

Section 4 — Genetic / Molecular Information

Causal gene. GET3 / ASNA1 (TRC40). NCBI Gene 439; 19p13.13; Ensembl ENSG00000198356; OMIM gene 601913.

Pathogenic variants (index family). Verhagen et al. 2019 identified **compound heterozygous** ASNA1/GET3 variants in two affected siblings:

Variant (cDNA)	Protein	Parental origin	Class	Functional consequence
c.913C>T	p.(Gln305*)	Paternal	Nonsense / premature termination codon	Decreased protein in myocardium and skin fibroblasts (null / LoF)
c.488T>C	p.(Val163Ala)	Maternal	Missense	Protein misfolding; less effective tail-anchored protein insertion (hypomorphic LoF)

"Exome sequencing was used to screen for the causative genetic defect in a pair of siblings with rapidly progressive dilated cardiomyopathy and death in early infancy." — [PMID: 31461301](#)

Variant classification. Both variants segregate with disease in a recessive pattern and are supported by functional evidence (reduced protein/insertion activity; failure to rescue zebrafish). They are consistent with pathogenic/likely-pathogenic classification under ACMG/AMP criteria (functional evidence PS3; recessive segregation; predicted null variant PVS1 for p.Gln305*).

Variant type/class. One nonsense/PTC allele (loss of protein) and one missense/misfolding allele. **Origin:** germline, biallelic (autosomal recessive). **Population frequency:** rare, consistent with an ultra-rare Mendelian disorder. **Functional consequence:** loss of function for both alleles (one null, one hypomorphic).

Modifier genes / epigenetics / chromosomal abnormalities. No CMD2H-specific modifier genes, epigenetic marks, or chromosomal abnormalities have been reported. This is a single-gene, small-variant disorder; no aneuploidy/translocation involvement is described.

Section 5 — Environmental Information

No environmental factors, lifestyle factors, or infectious agents are implicated in CMD2H. It is a purely genetic (Mendelian recessive) disorder. This contrasts with acquired/inflammatory DCM (e.g., viral myocarditis progressing to DCM [PMID: 42415986]; inflammatory cardiomyopathy [PMID: 41747776]), which is mechanistically distinct and not part of the CMD2H entity.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic loss-of-function variants in GET3/ASNA1** (e.g., p.Gln305 *null* + p.Val163Ala *misfolding*) **result in*** reduced GET3/ASNA1 protein level and/or ATPase/chaperone activity in cardiomyocytes.
2. Reduced GET3 activity **leads to** destabilization of the cytosolic pre-targeting complex and impaired recognition/delivery of tail-anchored (TA) membrane proteins to the ER membrane (GET/TRC pathway failure). (*Demonstrated in vitro via a TA-insertion assay for p.Val163Ala.*)
3. Failed TA-protein insertion **results in** reduced expression/mistargeting of multiple TA-protein substrates (SNAREs, ER/Golgi trafficking components, apoptosis regulators), **impairing** membrane trafficking and organelle proteostasis. (*Demonstrated in Asna1-KO mouse cardiomyocytes.*)
4. Disrupted cardiomyocyte proteostasis and membrane-protein homeostasis **lead to** a maladaptive compensatory transcriptional response (upregulation of protein-trafficking and Golgi-to-ER transport genes) and **impaired cardiomyocyte function/contractility**. (*Transcriptomics + zebrafish contractility data.*)
5. Cardiomyocyte dysfunction **results in** ventricular myocardial thinning (developmental) and/or ventricular dilation with reduced systolic function (postnatal/adult).
6. Progressive ventricular dilation and pump failure **lead to** heart failure and, in reported human cases, **death in early infancy**.

Steps 1–3 are directly demonstrated (human protein data, in vitro insertion assay, mouse molecular data). Steps 4–6 are supported by convergent zebrafish (contractility, lethality) and mouse (dilation, thinning, mortality) evidence; the precise identity of the critical cardiac TA-protein substrate(s) whose loss is rate-limiting for the heart phenotype remains **inferred rather than fully demonstrated**.

Detail by category

Molecular pathway — the GET/TRC (guided-entry of tail-anchored proteins) pathway.

GET3/ASNA1 (yeast Get3; mammalian TRC40) is the central cytosolic ATPase that binds the C-terminal transmembrane domain of TA proteins and delivers them to an ER membrane receptor formed by **WRB (=GET1)** and **CAML/CAMLG (=GET2)** (PMID: 21444755, PMID: 27226539). This is a post-translational, ATP-dependent targeting cycle.

Cellular processes. Membrane-protein biogenesis, ER-targeted protein trafficking, proteostasis, and (indirectly) apoptosis regulation (many apoptotic/BCL-2-family and SNARE proteins are TA proteins). ASNA1's ATPase activity is required for cell survival and ER homeostasis in other tissues (pancreatic progenitors [PMID: 29180572]; β -cell ER homeostasis [PMID: 26438609]), underscoring the pathway's general importance to proteostasis.

Protein dysfunction. p.Gln305* $\rightarrow$ truncation/loss of protein (reduced abundance in patient myocardium and fibroblasts). p.Val163Ala $\rightarrow$ misfolding and reduced TA-insertion efficiency (hypomorph). Net effect: loss of function.

"Protein expression was assessed in patient samples, followed by an in vitro tail-anchored protein insertion assay and functional analyses in zebrafish." — PMID: 31461301

Downstream molecular profiling (mouse). In cardiomyocyte-specific *Asna1*-KO mice, ASNA1 deficiency destabilized the pre-targeting complex and reduced expression of multiple TA-protein substrates, impairing membrane trafficking. Transcriptomics revealed compensatory (maladaptive) upregulation of protein-trafficking and Golgi-to-ER transport genes.

"ASNA1 deficiency destabilized the pre-targeting complex and reduced the expression of multiple TA protein substrates, impairing membrane trafficking and protein transport." — Feng et al. 2025, PMID: 41370295

Tissue-damage mechanisms. Impaired proteostasis → cardiomyocyte dysfunction → myocardial thinning (embryonic) or dilation and pathological remodeling (postnatal). Fibrosis/oxidative stress are downstream remodeling features common to DCM but not specifically characterized in CMD2H.

Upstream vs downstream. Upstream: GET3/ASNA1 LoF → GET/TRC pathway failure. Downstream: TA-substrate depletion → cardiomyocyte proteostatic stress → contractile dysfunction → ventricular dilation → heart failure.

Suggested ontology terms. - GO biological process: tail-anchored membrane protein insertion into ER membrane (GO:0071816); post-translational protein targeting to endoplasmic reticulum membrane (GO:0006620); protein insertion into ER membrane (GO:0045048). - GO cellular component: cytosol (GO:0005829); endoplasmic reticulum membrane (GO:0005789); TRC/GET complex. - CL cell type: cardiac muscle cell / cardiomyocyte (CL:0000746); regular ventricular cardiac myocyte (CL:2000046).

Section 7 — Anatomical Structures Affected

Organ level. Primary organ: **heart** (UBERON:0000948), specifically the **ventricular myocardium** — left ventricle (UBERON:0002084) and, in mouse, biventricular. Body system: **cardiovascular system** (UBERON:0004535). Secondary involvement: systemic congestion / heart-failure sequelae (pulmonary, hepatic, renal hypoperfusion) as in any severe infantile DCM.

Tissue and cell level. Tissue: **cardiac muscle tissue / myocardium** (UBERON:0002349). Target cell: **cardiomyocyte** (CL:0000746; ventricular cardiac myocyte CL:2000046). GET3/ASNA1 is broadly expressed, but the disease phenotype is myocardium-predominant.

Subcellular level. Compartments: **cytosol** (GO:0005829, site of GET3 targeting complex) and **ER membrane** (GO:0005789, destination of TA-protein insertion). Membrane trafficking machinery (Golgi-to-ER transport) is transcriptionally perturbed.

Localization / lateralization. Cardiac, bilateral/biventricular (mouse); left-ventricular predominant description in human DCM. No focal or asymmetric lateralization is characteristic.

Section 8 — Temporal Development

Onset. Congenital/infantile. In the index human family, disease was evident and rapidly progressive in early infancy. In constitutive *Asna1*-KO mice, myocardial thinning appears by embryonic day 16.5 with perinatal lethality, indicating a developmental component.

Progression. Rapid. Human course: rapidly progressive DCM with death in early infancy. Inducible adult-cardiomyocyte *Asna1* deletion in mice caused rapid ventricular dilation, impaired function, remodeling, and early mortality — indicating that GET3/ASNA1 is required for both cardiac development and ongoing adult cardiomyocyte maintenance.

"Constitutive *Asna1* deletion during embryogenesis caused perinatal lethality with marked ventricular myocardial thinning by embryonic day 16.5, whereas inducible deletion in adult cardiomyocytes led to rapid ventricular dilation, impaired cardiac function, pathological remodeling, and early mortality." — Feng et al. 2025, [PMID: 41370295](#)

Course pattern. Progressive, non-remitting, fatal in reported cases. No spontaneous remission described. **Critical period:** perinatal/infancy is the window of vulnerability; there is no established intervention window beyond supportive heart-failure care and transplantation.

Section 9 — Inheritance and Population

Inheritance. Autosomal recessive (biallelic GET3/ASNA1 LoF). Heterozygous carriers are unaffected — consistent with gnomAD constraint showing tolerance of a single LoF allele.

Penetrance / expressivity. In the single reported family both biallelic siblings were severely affected (apparently high penetrance for the severe biallelic genotype), but the extremely small case count precludes robust penetrance/expressivity estimates.

Epidemiology. Ultra-rare. Only a single index family (two siblings) is definitively reported for CMD2H, plus supporting functional data; prevalence/incidence are not established. For context, idiopathic DCM overall has an approximate historical prevalence near 1:2500 and is "the most common cardiomyopathy worldwide" ([PMID: 42511685](#)), but CMD2H represents a vanishingly small monogenic subset.

Carrier frequency / founder effects / consanguinity. Not established; no founder effect reported. **Sex ratio:** no sex bias expected for an autosomal recessive gene; not established given tiny sample. **Geographic distribution:** not established.

gnomAD constraint (ENSG00000198356; chr19:12,737,139–12,748,323, GRCh38): pLI = 0.05; LoF o/e = 15/32.4 (oe_lof = 0.46, 90% CI 0.31–0.71); lof_z = 2.59; mis_z = 2.51 — supports a recessive, loss-of-function model with healthy heterozygous carriers.

Section 10 — Diagnostics

Genetic testing is the definitive diagnostic modality. Because CMD2H is clinically indistinguishable from other infantile DCM on imaging alone, molecular diagnosis rests on identifying biallelic GET3/ASNA1 variants.

- **Recommended approach:** trio exome sequencing (WES) or genome sequencing (WGS), or a comprehensive cardiomyopathy gene panel that includes ASNA1/GET3, with segregation testing of parents to confirm compound heterozygosity/biallelic status. Exome sequencing was the method that discovered the disease gene.
- **Single-gene testing:** targeted ASNA1/GET3 sequencing (with deletion/duplication analysis) is appropriate once the diagnosis is suspected or in cascade testing.
- **Variant interpretation:** classify per ACMG/AMP using ClinVar/ClinGen, incorporating functional evidence (reduced protein, impaired TA-insertion) and recessive segregation.

Clinical/imaging tests (phenotype confirmation, non-specific). Echocardiography and cardiac MRI to document ventricular dilation and reduced ejection fraction; ECG; natriuretic peptides (BNP/NT-proBNP) and troponin as heart-failure biomarkers. Endomyocardial biopsy could show reduced GET3 protein (as in the index case) but is not routinely required.

Differential diagnosis. Other genetic infantile/pediatric DCMs (e.g., TTN, LMNA, sarcomeric and metabolic cardiomyopathies), CTNNA3-related recessive DCM [PMID: 42471840], GATAD1-related CMD2B (see disambiguation), mitochondrial/metabolic cardiomyopathies, and acquired causes (viral myocarditis, inflammatory cardiomyopathy). Genetic testing distinguishes CMD2H.

Screening. Cascade genetic testing of at-risk relatives and carrier testing of parents for reproductive counseling. No population newborn screening exists for this ultra-rare disorder.

Section 11 — Outcome / Prognosis

Prognosis is poor in reported cases — rapidly progressive DCM with death in early infancy in the two index siblings. Mouse models mirror this severity (perinatal lethality with constitutive KO; early mortality with adult inducible KO).

Survival / mortality. Reported human outcome: fatal in infancy. Definitive survival statistics are not available due to the tiny case count.

Morbidity / complications. Severe heart failure, arrhythmia risk, and the sequelae of low cardiac output. For DCM broadly, non-ischaemic DCM confers ~12-fold higher out-of-hospital cardiac arrest incidence versus the general population (532 vs 45 per 100,000 person-years) with higher shockable-rhythm rates [PMID: 42536775] — underscoring arrhythmic risk that informs management of any DCM, though CMD2H's infantile lethality dominates its natural history.

"The incident rate of OHCA was approximately 12 times higher in NIDCM patients compared with the general population (incidence rate 532 vs 45 per 100 000 person years)." — [PMID: 42536775](#)

Prognostic factors. Biallelic null genotype and infantile onset predict severe outcome. No CMD2H-specific prognostic biomarkers are established.

Section 12 — Treatment

There is no disease-specific or targeted therapy for CMD2H. Management follows **standard guideline-directed heart-failure therapy** for pediatric/infantile DCM, with **heart transplantation** as the definitive option for end-stage disease.

Pharmacotherapy (supportive, standard HF care; NCIT terms in brackets). - Diuretics for congestion (e.g., furosemide) [Loop Diuretic]. - ACE inhibitors / ARBs / ARNI for afterload reduction and remodeling [Angiotensin-Converting Enzyme Inhibitor, NCIT:C776]. - Beta-blockers where tolerated [Beta-Adrenergic Blocker, NCIT:C2568]. - Mineralocorticoid receptor antagonists. - SGLT2 inhibitors are now standard in adult HFrEF; pediatric applicability is individualized. - Inotropic support for acute decompensation.

Device / advanced therapies. ICD for arrhythmic protection in appropriate patients; mechanical circulatory support (VAD) as a bridge; **cardiac transplantation** [Heart Transplantation, NCIT:C15326] for refractory disease.

Advanced / experimental therapeutics. No approved gene therapy, RNA therapy, or targeted small molecule exists for GET3/ASNA1-related DCM. Because the mechanism is recessive loss of function, **gene-replacement / gene-addition strategies (e.g., AAV-delivered ASNA1)** are conceptually rational future directions but are entirely preclinical/hypothetical at present.

Pharmacogenomics. None specific to CMD2H.

Section 13 — Prevention

Primary prevention of the disease itself is genetic: **carrier identification and reproductive counseling**. Options for at-risk couples (both carriers) include prenatal diagnosis and **preimplantation genetic testing (PGT-M)**.

- **Genetic counseling** for families with an affected child: 25% recurrence risk per pregnancy for two carrier parents (autosomal recessive).
- **Cascade carrier testing** of relatives.
- **Secondary/tertiary prevention** for an affected infant is limited to early heart-failure management, arrhythmia surveillance/ICD where appropriate, and timely transplant evaluation.

No immunization, behavioral, or public-health prevention applies to this monogenic disorder.

Section 14 — Other Species / Natural Disease

Evolutionary conservation. GET3/ASNA1 is a highly conserved ATPase. Functional orthologs span yeast (**GET3**), *C. elegans* (**asna-1**), zebrafish (**asna1**), and mouse (**Asna1**), all participating in TA-protein insertion. This deep conservation supports the mechanistic model and enables cross-species modeling.

"Our findings point toward a critical role of the tail-anchored membrane protein insertion pathway in vertebrate cardiac function and disease." — Verhagen et al. 2019, [PMID: 31461301](#)

Orthologous genes (suggested identifiers to populate): mouse *Asna1*, zebrafish *asna1*, *C. elegans* *asna-1*, *S. cerevisiae* GET3. **Natural disease in companion animals/wildlife:** no OMIA entry for spontaneous ASNA1/GET3 cardiomyopathy is established; the animal data are experimental models rather than naturally occurring disease. **Zoonotic potential:** not applicable (genetic disorder).

Section 15 — Model Organisms

Zebrafish (*Danio rerio*, *asna1*). Loss of *asna1* produced reduced cardiac contractility and early lethality; wild-type *asna1* mRNA rescued the phenotype, whereas mRNA carrying either patient variant (p.Gln305* or p.Val163Ala) failed to rescue — establishing both human alleles as functionally deleterious and validating the gene–disease relationship in vivo.

Mouse (*Mus musculus*, *Asna1*) — cardiomyocyte-specific conditional knockouts (Feng et al. 2025, [PMID: 41370295](#)). - *Constitutive* cardiomyocyte deletion → perinatal lethality with ventricular myocardial thinning by E16.5. - *Inducible* adult deletion → rapid ventricular dilation, impaired cardiac function, pathological remodeling, and early mortality. - *Molecular:* destabilized pre-targeting complex, reduced TA-substrate expression, impaired membrane trafficking; compensatory transcriptional upregulation of trafficking genes.

Phenotype recapitulation. Excellent — mouse models reproduce the core human features (ventricular dilation, impaired function, early death) and provide the mechanistic link to TA-protein proteostasis. Zebrafish provide orthogonal in vivo validation and a variant-specific rescue assay.

Model limitations. Species differences in cardiac physiology; conditional models are engineered LoF rather than the exact human compound-heterozygous genotype; human natural-history and therapeutic-response data remain minimal.

Suggested resources: MGI (*Asna1*), ZFIN (*asna1*), IMPC/IMSR for mouse alleles.

Key Findings (with evidence)

Finding 1 — CMD2H is caused by biallelic GET3/ASNA1, not GATAD1 (critical disambiguation)

MONDO:0859358 "cardiomyopathy, dilated, 2H" cross-references OMIM:620203, MedGen/UMLS C5774296, and GARD 0026714 (verified via Monarch/OLS4). The causal gene is **GET3/ASNA1** (NCBI Gene 439, 19p13.13), whose official aliases include ASNA1, TRC40, ARSA1, and **CMD2H**. Verhagen et al. 2019 identified compound heterozygous ASNA1 variants in two siblings with fatal infantile DCM. **GATAD1 causes a distinct disease, CMD2B (OMIM 614672)** — the initial GATAD1 leads in this investigation belong to CMD2B and do not apply to CMD2H.

"Biallelic variants in ASNA1 cause severe pediatric cardiomyopathy and early death." — [PMID: 31461301](#)

Finding 2 — Pathogenic variant spectrum: a truncating null plus a misfolding missense, both LoF

Paternal c.913C>T, p.(Gln305*) (null; reduced protein in myocardium/fibroblasts) + maternal c.488T>C, p.(Val163Ala) (misfolding; impaired TA insertion). Both are loss-of-function; zebrafish rescue failed for both. Germline, biallelic, autosomal recessive.

"Exome sequencing was used to screen for the causative genetic defect in a pair of siblings with rapidly progressive dilated cardiomyopathy and death in early infancy." — [PMID: 31461301](#)

Finding 3 — Mechanism: defective GET/TRC tail-anchored-protein insertion impairs cardiomyocyte function

GET3/ASNA1 (TRC40) is the cytosolic ATPase delivering TA proteins to the ER via the WRB(GET1)/CAML(GET2) receptor. Biallelic LoF reduces TA-insertion capacity (in vitro assay), reducing cardiac contractility (zebrafish) and causing dilation/failure.

"Protein expression was assessed in patient samples, followed by an in vitro tail-anchored protein insertion assay and functional analyses in zebrafish." — [PMID: 31461301](#)

Finding 4 — Mouse cardiomyocyte-specific Asna1 KO recapitulates CMD2H

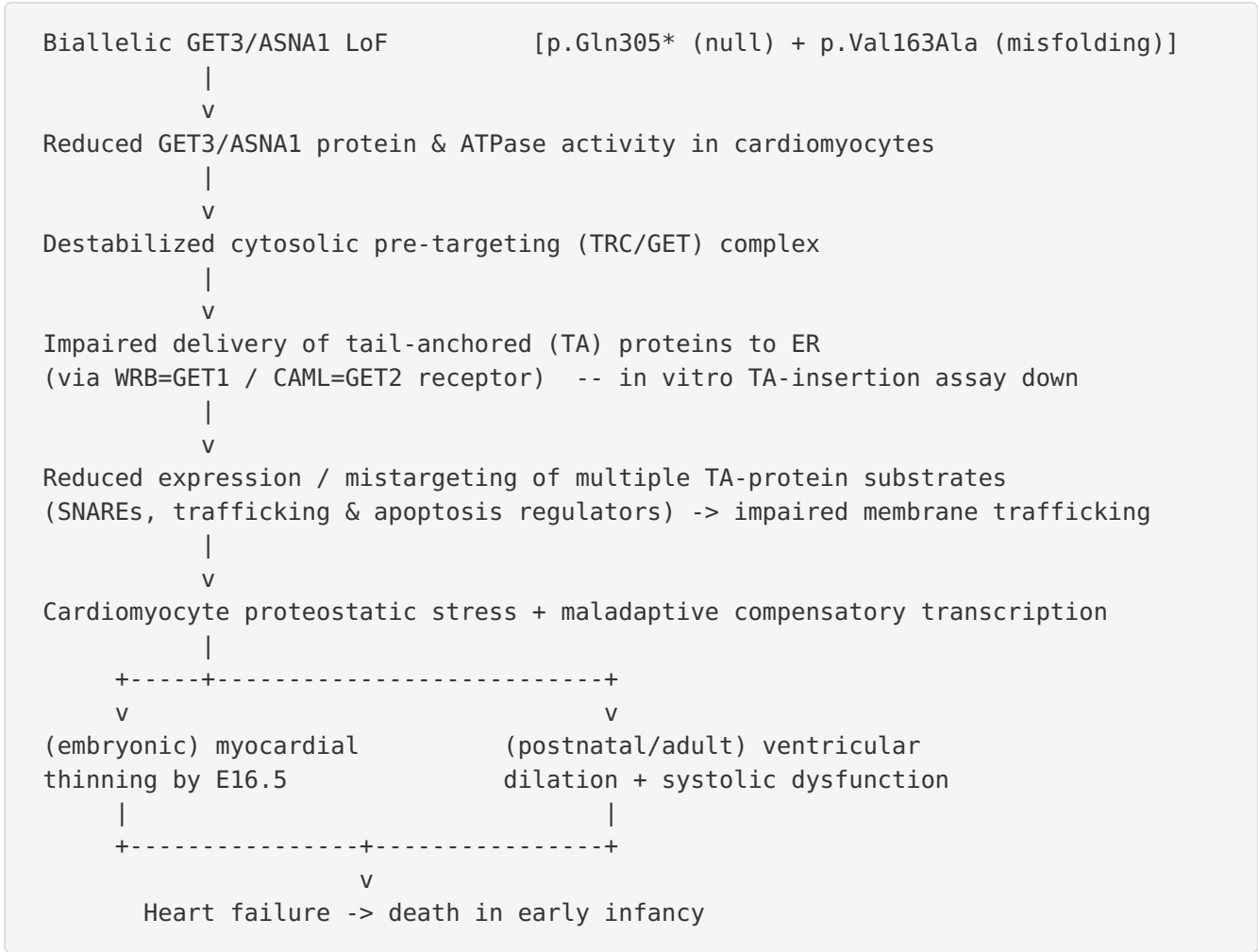
Constitutive KO → perinatal lethality + myocardial thinning by E16.5; inducible adult KO → ventricular dilation, impaired function, remodeling, early death; molecular destabilization of the pre-targeting complex with reduced TA-substrate expression.

"ASNA1 deficiency destabilized the pre-targeting complex and reduced the expression of multiple TA protein substrates, impairing membrane trafficking and protein transport." — [PMID: 41370295](#)

Finding 5 — gnomAD constraint supports a recessive LoF model

pLI = 0.05; oe_lof = 0.46 (90% CI 0.31–0.71); lof_z = 2.59; mis_z = 2.51 — constrained but heterozygous-LoF-tolerant, consistent with unaffected carriers and biallelic disease.

Mechanistic Model / Interpretation



Upstream driver: GET/TRC pathway failure. Downstream effectors: TA-substrate depletion → cardiomyocyte dysfunction → dilation → pump failure. The identity of the single most rate-limiting cardiac TA substrate remains inferred.

Evidence Base

PMID	Title (abbrev.)	Role	Support / Challenge
31461301	<i>Biallelic Variants in ASNA1...</i> (Verhagen 2019)	Index human family; gene discovery; in vitro + zebrafish	Core support — establishes GET3/ASNA1 causation, variants, mechanism
41370295	Cardiomyocyte-specific Asna1 KO mice (Feng 2025)	Mouse model	Core support — recapitulates dilation/thinning; molecular mechanism
Monarch / OLS4	MONDO:0859358 ↔ OMIM:620203 ↔ GET3/ASNA1	Identifier verification	Core support — disease/gene mapping; disambiguation from CMD2B
gnomAD (ENSG00000198356)	Constraint metrics	Population genetics	Support — recessive LoF model
21444755 , 27226539	WRB/CAML TA-insertion receptor	Pathway biology	Support — GET/TRC receptor machinery
29180572 , 26438609	ASNA1 in pancreas / β -cell ER homeostasis	Pathway importance	Support — ASNA1 ATPase essential for survival/proteostasis
42536775	OHCA risk in non-ischaemic DCM	Clinical context	Context — DCM arrhythmic risk (broad)
42511685	Geographical/ethnic heterogeneity in genetic DCM	Epidemiology context	Context — DCM epidemiology framing
41898170	Emerging mechanisms/therapies in DCM	Therapy context	Context — general DCM management
21965549	GATAD1 in AR DCM (Theis 2011)	Disambiguation	Belongs to CMD2B , not CMD2H

Limitations and Knowledge Gaps

1. **Extremely small human evidence base.** The core clinical description rests on a single family (two siblings). Prevalence, penetrance, expressivity, carrier frequency, sex ratio, and geographic/ethnic distribution are essentially unknown.
2. **Substrate specificity unresolved.** It is not established which cardiac TA-protein substrate(s) are most critically depleted; the link from "reduced TA insertion" to "cardiomyocyte failure" is mechanistically plausible and model-supported but the rate-limiting node is inferred.
3. **No natural-history or therapeutic-response data** specific to CMD2H; treatment recommendations are extrapolated from general DCM/heart-failure care.
4. **Historical disambiguation risk.** Much online and literature content conflates "Dilated Cardiomyopathy 2H" with GATAD1; this report explicitly corrects that — CMD2H = GET3/ASNA1; GATAD1 = CMD2B.
5. **No CMD2H-specific biomarkers, modifiers, or epigenetic data** are available.

Proposed Follow-up Experiments / Actions

1. **Expand the case series.** Query GeneMatcher, DECIPHER, and large cardiomyopathy registries (e.g., the Dutch Cardiomyopathy Registry [PMID: 42573902]) for additional biallelic ASNA1/GET3 patients to define phenotypic range and penetrance.
2. **Define the critical cardiac TA-substrate(s).** Proteomics/TA-substrate profiling in Asna1-KO cardiomyocytes and patient-derived iPSC-cardiomyocytes to pinpoint the rate-limiting insertion defect.
3. **iPSC-CM disease modeling.** Generate patient-derived iPSC-cardiomyocytes (and isogenic corrected controls) to characterize contractility, ER stress, and trafficking, and to serve as a therapeutic-screening platform.
4. **Genotype-phenotype and rescue studies.** Systematic in vitro insertion assays across additional ASNA1 variants to build an ACMG-grade functional evidence base (PS3) for clinical variant interpretation.
5. **Preclinical gene-replacement proof of concept.** Test AAV-mediated ASNA1 restoration in the inducible Asna1-KO mouse as a rational strategy for a recessive LoF disorder.

6. **Curate identifiers.** Ensure knowledge-base entries link MONDO:0859358 → OMIM 620203 → GET3/ASNA1 (gene OMIM 601913) and flag the GATAD1/CMD2B distinction to prevent propagation of the historical error.

Evidence source legend: human clinical (Verhagen 2019, index family), in vitro (TA-insertion assay), model organism (zebrafish asna1; mouse Asna1 conditional KO), computational/population genetics (gnomAD constraint), and database/ontology cross-referencing (Monarch/OLS4, OMIM, MONDO, NCBI Gene).

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